

LESSON 2-3 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find Quartiles and IQR

Builds on Lesson 2-5 Session 1 • 6.SP.B.5c

TODAY'S LESSON

In problem 2 you find the median, Q1, Q3, and IQR of each carrot plot.

YOU NEED TOFind Q1, Q3, and the IQR of a data set. **You will use it in Practice Form A, problems 1 and 2.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Order** the data. Find the median.
- 2** Find the **quartiles**: the middle of each half.
- 3 IQR** = $Q3 - Q1$.

WATCH ME*the steps in action***Find the IQR of 12, 15, 17, 20, 22, 26, 30.****Step 1** The median is 20.**Step 2** Lower half 12, 15, 17: $Q1 = 15$. Upper half 22, 26, 30: $Q3 = 26$.**Step 3** $IQR = 26 - 15 = 11$.**Answer 11****TRY IT WITH HELP***fill in each step*

- 1.**
- Find the IQR of 5, 8, 12, 14, 16, 20, 25.

Step 1 The median is _____.**Step 2** $Q1 =$ _____. $Q3 =$ _____.**Step 3** $IQR =$ _____.

- 2.**
- Find the IQR of 2, 4, 6, 8, 10, 12, 14, 16.

Step 1 The median is _____.**Step 2** $Q1 =$ _____. $Q3 =$ _____.**Step 3** $IQR =$ _____.**YOUR TURN***use the steps on your own*

- 3.**
- Each list is in order. Find the IQR.

a. 1, 3, 5, 7, 9, 11, 13 ____**b.** 10, 14, 15, 18, 20, 25, 30 ____**c.** 40, 42, 45, 50, 52, 58, 60 ____**d.** 6, 7, 9, 12, 13, 15 ____

Compare Spread With IQR

Builds on Lesson 2-5 Session 2 • 6.SP.B.5c

TODAY'S LESSON

In problem 6 the coach wants the team with the smaller IQR. It scores closer to its usual total.

YOU NEED TO

Use the IQR to tell which group is more consistent. **You will use it in Practice Form A, problems 6 and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** Find each group's **IQR**.
- 2** **Compare** the IQRs.
- 3** The **smaller** IQR is more consistent.

WATCH ME*the steps in action*

Class P has an IQR of 14. Class Q has an IQR of 6. Which is more consistent?

Step 1 P: 14. Q: 6.

Step 2 $6 < 14$.

Step 3 Q has the smaller IQR.

Answer **Class Q**

TRY IT WITH HELP*fill in each step*

4. Team 1: $Q1 = 26$, $Q3 = 34$. Team 2: $Q1 = 15$, $Q3 = 40$.

Step 1 IQR 1 = _____. IQR 2 = _____.

Step 2 _____ < _____

Step 3 More consistent: Team _____.

5. Store A has an IQR of 21. Store B has an IQR of 9.

Step 1 A: _____. B: _____.

Step 2 _____ < _____

Step 3 More consistent: Store _____.

YOUR TURN*use the steps on your own*

6. Write the IQR of the more consistent set.

a. IQR 9 or 15: ____

b. IQR 20 or 12: ____

c. IQR 4 or 7: ____

d. IQR 30 or 31: ____

e. IQR 18 or 6: ____

f. IQR 11 or 25: ____

READY CHECK*a problem like the one in today's lesson*

7. Which IQR shows the most consistent data?

A IQR 15

B IQR 3

C IQR 9

D IQR 22

CHECK YOURSELF
☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____